Enterprise Web Software Development

COMP1640

EWSD Group_B1 Report

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University of Greenwich
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2. Introduction

This report will illustrate how Group B1 collaborated to create a website that showcased the best contributions of students from the university over the year. By reading this report, you will be able to understand the functions implemented by Group B1 in the system, the role-based features of the web platform, and its usage. (Gurnov, 2025)

The Purpose of the system

The university wants to implement a system that will highlight the best contributions of its students during the 2026-2027 academic year. Furthermore, they aim to include actions from other responsible parties that influence students' articles to ensure that the system runs with the initial requirements. Then, the university has 4 faculties: Science, Engineering, Art and Humanities, Medicine and Business. (Gurnov, 2025)

Therefore, the ultimate goal of the system is to allow students in each department to submit contributions and implement the actions of other responsible people. Provides a summary of the functions based on user roles, as outlined below. (Gurnov, 2025)

User roles and Functionalities

The typical users within the platform are students, marketing manager, marketing coordinators, administrators, and guests. Therefore, the functions within the system differ depending on the role of each user, as summarized below. (Gurnov, 2025)

Student: As a student, he/she will have a registered account that is generated by the admin before accessing the system. Then, he/she will need to log in with your credentials. Once he/she is logged into the platform, the person (as a student) will be able to submit one or more contributions or high-quality images. However, before submitting the articles, student must agree to the site policies within the app set by the university. Items can be complied with before the closure date and updated before the final date. (Gurnov, 2025)

Marketing Coordinator: As a marketing coordinator, the person will need to log into the system using the credentials assigned by the administration to his/her faculty. After logging in, he/she will be able to read, review, and make decisions (select, request revisions, or reject) on contributions uploaded by students within his/her department. Feedback should be provided to students within 14 days. (Gurnov, 2025)

Marketing Manager: The manager is able to view articles selected by marketing coordinators across faculties and download them in ZIP file format. (Gurnov, 2025)

Administrator: Admin can define the final selection date of articles and …….

(Gurnov, 2025)

Guest: If a person logs in to the "EchoPress" website as a guest, he/she will be able to view selected reports in the department associated with that account. (Gurnov, 2025)

Business point of view

From a business perspective, the team (Group B1) built the platform to meet the high-level needs of the university in a short period of time. Therefore, we believe we have overcome obstacles that could have lowered business standards, such as implementation delays and reduced profitability.

On the other hand, the university will also be able to easily involve their students in the campaign and implement their activities with less difficulty. Nevertheless, inadequate system management may result in unwanted complications and potentially harm the institution's reputation.

(Gurnov, 2025)

As summary, the above statements show that the team has built the web platform in accordance with the requirements that should be included. Therefore, it is believed that the system will be able to effectively implement the necessary functions of the university. (Gurnov, 2025)

3. Agile Documentation

3.1 Meeting Notes

Sprint 0

No.	Week	Minutes	Attendance	Discussion	Tasks for next week
1.	1 / 28 / 2026	8:30 – 9:30Am 9:30 –10:30Am	Attended All	1. The first stage was diverting roles for each individual. 2. Discussed about API. 3. Consulted for frame-work. 4. Conversed about database usage. 5. Researched test plans. 6. Created repository for common work place.	1. Preparing about the database. 2. Researching for hosting. 3. Assign and design about user management.

No.	Week	Minutes	Attendance	Discussion	Tasks for next week
2.	2 / 8 / 2026	8:30 – 9:30Am 9:30 –10:30Am	Attended All	1. Discussed on Role and Responsibilities. 2. Examined the materials of course work. 3. Explained about the wireframe. 4. Analysed the RBAC of platform.	1. Preparing about Wireframe. 2. Initiating to the front-end development. 3. Initiating to the back-end development.

				5. Predicted the evaluation issues. 6. Examined additional features and research their high-level requirements.	4. Make ready for ERD and UI.

No.	Week	Minutes	Attendance	Discussion	Tasks for next week
3.	2 / 18 / 2026	9:30 –10:30Am	Attended All	1. Finalized the tools and requirements for UML and Wireframe. 2. Discussed the file location within database. 3. Define and agreement on requirements of the project. 4. Define the steps to work done. 5. Preparing the needs to initiate the sprint 1. (project development) 6. Outlined the works that will consist in sprint 1.	1. Have to discuss and reveals the problems within project setup. 2. Examine and analyze the functionalities of student role.

Total minutes of **Sprint 0**: 300 minutes.

Sprint 1

No.	Week	Minutes	Attendance	Discussion	Tasks for next week
1.	2 / 25 / 2026	9:30 Am – 10:30 Am 10:45 Am – 11:00 Am	Attended All	1. Discussed the identity verification process and initiated the procedure. (Include token management, login/register flow, password reset, and security vulnerabilities). 2. Database Foundation. 3. Examined the relationship between the tables and the initial data. 4. Initiated user stories. 5. UI/UX component alignment. 6. Discussed on RBAC within the web platform. 7. Identified and specified the work items to be accomplished during Sprint 1.	1. Preparing for product backlogs. 2. Preparing for student dashboard. (Functionality). 3. Access Control. 4. Data upload in multiple time. 5. Rules and regulation validation.

Total minutes of **Sprint 1**: 75 minutes.

Sprint 2

No.	Week	Minutes	Attendance	Discussion	Tasks for next week
1.	3 / 4 / 2026	9:30 Am – 10:30 Am	Attended All	1. Checking whether that the works from Sprint 1 has been finished. (or) Any backlogs. 2. File type validation. (Submission process) 3. Discussed file size limitation process. 4. Navigation mechanism and performance on student dashboard. 5. Presented data accuracy on student dashboard. 6. Version control. (T&C validation) 7. Discussed on blocking method. (T&C validation) 8. Information architecture. 9. Identified and specified the work items to be accomplished during Sprint 2.	1. Preparing for product backlogs. 2. Preparing for marketing coordinator's functionalities. 3. Initiate for email-notification process. 4. Examine and analyse to begin the commenting system. 5. Preparing for that will consist inside the Sprint 3.

Total minutes of **Sprint 2**: 60 minutes.

Sprint 3

No.	Week	Minutes	Attendance	Discussion	Tasks for next week
1.	3 / 11 / 2026	9:30 Am – 10:30 Am	Attended All	1. Checking whether that the works from Sprint 2 has been finished. (or) Any backlogs. 2. Addressed the ongoing works in UI development. 3. Initiated to connect the front end with backend. 4. Discuss the strategies to test the **User manual** part. 5. Presented the group about modules that were fully implemented in backend code, and describe which parts are ahead to build. 6. Re-examine on assumption. 7. Analyse the works of sprint-3 by using its gen chart. 8. Review on front end development. 9. Identified and specified the work items to be accomplished during Sprint 3.	1. Preparing for product backlogs. 2. Start-up of front-end design. 3. Examine and analyse to begin the commenting system. 5. Preparing for that will consist inside the Sprint 4.

Total minute of **Sprint 3**: 60 minutes.

Sprint 4

No.	Week	Minutes	Attendance	Discussion	Tasks for next week
1.	3 / 18 / 2026	9:30 Am – 10:30 Am 10:30 Am – 11:00 Am	Attended All	1. Checking whether that the works from Sprint 3 has been finished. (or) Any backlogs. 2. Presented the front end of all the roles within the platform to ensure collective awareness and understanding. 3. Contributions filtering function on guest and coordinator accounts. 4. Discuss how many reports should be inside the system. 5. The blueprint for the reporting system was explained and members shared their ideas. 6. Developer reminded front end designer to send the UI design and specified data within the front end process. 7. Analysed the changes that could occur in the database.	1. Preparing for product backlogs. 2. Preparing for marketing coordinator's functionalities. 3. Initiate for email-notification process. 4. Examine and analyse to begin the commenting system. 5. Preparing for that will consist inside the Sprint 4.

				8. Identified and specified the work items to be accomplished during Sprint 3.	

Total minute of **Sprint 4**: 75 minutes.

3.2 User Stories

Marketing Manager

1. As a marketing manager, he/she want to access the platform to oversee the processes running within the system.

2. As a marketing manager, he/she want his/her credentials to be hashed using the Argon2id algorithm so that their identity remains protected against brute-force and rainbow table attacks.

3. As a marketing manager, he/she would like to see the contributions selected from each faculty.

4. As a marketing manager, he/she would like to download selected articles from each department as a zip file and save it on my computer after the final closure date.

5. As a marketing manager, he/she want to see how many students are registered in each faculty.

Marketing Coordinator

1. As a marketing coordinator, he/she want to access the system to monitor the submissions of students in my department.

2. As a marketing coordinator, he/she want his/her credentials to be hashed using the Argon2id algorithm so that their identity remains protected against brute-force and rainbow table attacks.

3. As a marketing coordinator, he/she want to receive an email notification after each submission so that I can comment on the manuscript within 14 days.

4. As a marketing coordinator, he/she would like to provide feedback (select, reject, or need revision) on student contributions.

5. As a marketing coordinator, he/she would like to have a statically page that displays the articles submitted by students within my institution to ensure effective connection.

6. As a marketing coordinator, he/she want to observe how many students are registered inside my respective faculty.

7. As a marketing coordinator, he/she require a dedicated page that describes the total number of contributions in my department that have been selected, rejected, or notified for editing.

Administrator

1. As an administrator, he/she want a dedicated logging structure so that he/she can track user login activity and browser types as required by the V2 project specifications.

2. As an administrator, he/she want his/her credentials to be hashed using the Argon2id algorithm so that their identity remains protected against brute-force and rainbow table attacks.

3. As an administrator, he/she need to have special access within the platform to be able to modify the closure dates.

4. As an administrator, he/she require privileged access for monitoring and managing users inside the web-based application.

5. As an admin, he/she want to access the statistics page to understand the system's performance and usability.

Student

1. As a student, he/she want to register my credentials in the system so that his/her can be recognized as an authorized student within the university's web application.

2. As a student, he/she want his/her credentials to be hashed using the Argon2id algorithm so that their identity remains protected against brute-force and rainbow table attacks.

3. As a student, he/she would like to log into the system to post his/her thoughts and opinions (or) pictures.

4. As a student, he/she would like to confirm that my manuscript complies with the established rules and regulations before submitting article.

5. As a student, he/she want to revise submitted contributions before the final closure date.

6. As a student, he/she wish to receive timely feedback from the marketing coordinator in his/her respective department.

7. As a student, he/she want to organize my ideas into categories and groups so that it is easier to conduct research later.

8. As a student, he/she would like to have a statistics page where I can see my selected articles, articles that he/she want to edit, and articles that got rejected for.

Guest

1. As a guest, he/she would like to access the web-platform to view selected contributions from his/her respective institute.

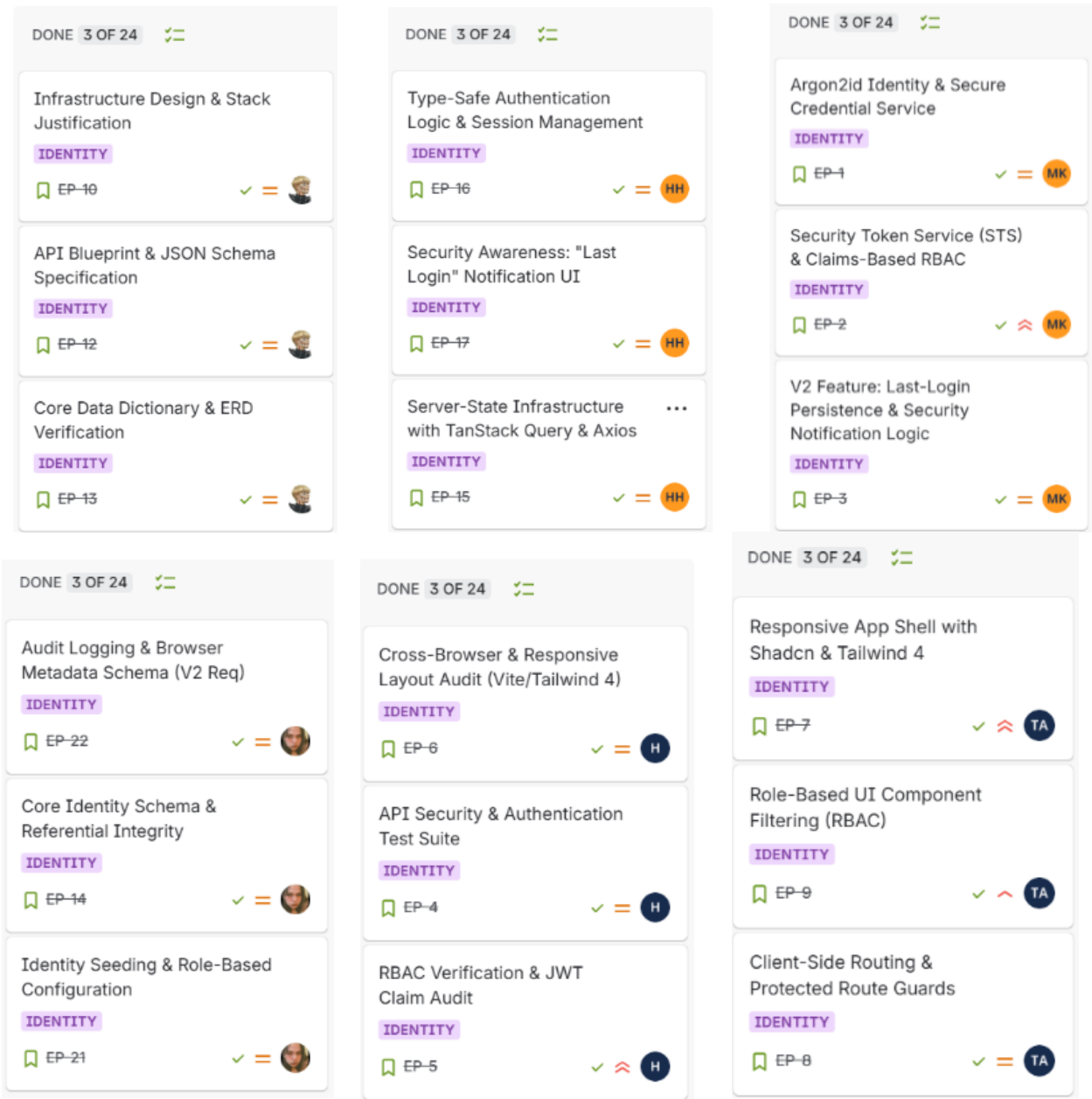
2. As a guest, he/she want his/her credentials to be hashed using the Argon2id algorithm so that their identity remains protected against brute-force and rainbow table attacks.

3.5 Sprints of the project

Then, the team needs to divide the tasks among each member to ensure that the product is delivered on time. Only then can the members fully focus on the sprint goals that they should be working on and achieve the specified results within a given time frame.

The team decided to use a tool within the development process to track iteration targets and record remaining or incomplete tasks. The final result was Jira, a project management tool supported by Atlassian. Jira is responsive on both desktop and mobile, making it easy for team members to use. The contributions of each team member within each sprint are outlined below.

In sprint 1



DONE 3 OF 24

Greenwich Digital Identity & Tailwind 4 Design Tokens

IDENTITY

~~EP-18~~ YO

High-Fidelity App Shell & Login Flow

IDENTITY

~~EP-20~~ YO

Atomic Component Library (Shadcn Customization)

IDENTITY

~~EP-19~~ YO

DONE 3 OF 24

Agile Ceremony Documentation & Minutes

IDENTITY

~~EP-23~~

Core Information Architecture (Sitemap)

IDENTITY

~~EP-25~~

Group Repository Setup & Secure Access Management

IDENTITY

~~EP-24~~

In Sprint 2

DONE 3 OF 24

- UML Usecase Diagram (The Submission Life Cycle) — CONTENT — ~~EP-27~~
- Updated Data Dictionary & Information Architecture — CONTENT — ~~EP-26~~
- File Storage Rationale & Security Brief — CONTENT — ~~EP-28~~

DONE 3 OF 24

- Sprint 2 Burndown & Velocity Tracking — CONTENT — ~~EP-50~~
- Weekly Risk Register Update (Week 2) — CONTENT — ~~EP-49~~
- Sprint 2 Planning Minutes & Goal Setting — CONTENT — ~~EP-48~~

DONE 3 OF 24

- Cross-Device Mobile Responsiveness Audit — CONTENT — ~~EP-41~~ — H
- T&C Agreement Persistence Verification — CONTENT — ~~EP-40~~ — H
- Multipart/Form-Data Boundary & Size Testing — CONTENT — ~~EP-39~~ — H

DONE 3 OF 24

- Submission History Data Fetching (useQuery) — CONTENT — ~~EP-32~~ — HH
- Client-Side File Validation & Preview Logic — CONTENT — ~~EP-31~~ — HH
- TanStack Mutation for Multi-Asset Submission — CONTENT — ~~EP-30~~ — HH

DONE 3 OF 24

- Student Submission History API — CONTENT — ~~EP-35~~ — MK
- Secure File Storage Service — CONTENT — ~~EP-34~~ — MK
- Multi-Asset Upload Controller (.NET 8) — CONTENT — ~~EP-33~~ — MK

DONE 3 OF 24

- T&C Agreement Audit Table — CONTENT — ~~EP-37~~
- Submission Metadata Stored Procedures/Views — CONTENT — ~~EP-38~~
- Contribution & Asset Schema — CONTENT — ~~EP-36~~

DONE 3 OF 24

Responsive File Selection UI
CONTENT
~~EP-44~~ TA

Mandatory T&C Agreement Modal
CONTENT
~~EP-43~~ TA

Student Dashboard Navigation
CONTENT
~~EP-42~~ TA

DONE 3 OF 24

Mobile-First "My Submissions" Dashboard
CONTENT
~~EP-47~~ YO

Student Submission Wizard & File Upload UX
CONTENT
~~EP-46~~ YO

Terms & Conditions (T&C) Agreement Experience
CONTENT
EP-45 YO

In Sprint 3

DONE 3 OF 27

- User Manual: Coordinator Role — COORDINATION — ~~EP-75~~
- UML Sequence Diagram (Coordination & Feedback) — COORDINATION — ~~EP-74~~
- Exception Report Logic Documentation — COORDINATION — ~~EP-76~~

DONE 3 OF 27

- Cross-Faculty Security Audit — COORDINATION — ~~EP-72~~ — H
- Logic Testing & Notification Flow 14-Day Deadline Logic Test — COORDINATION — ~~EP-71~~ — H
- Email Delivery & Link Verification — COORDINATION — ~~EP-73~~ — H

DONE 6 OF 27

- Integration & Business Logic Verification Faculty-Gated Route Protection — COORDINATION — ~~EP-68~~ — HH
- Final Coordinator UI Polish — COORDINATION — ~~EP-70~~ — HH
- Notification Polling/Status Check — COORDINATION — ~~EP-67~~ — HH

DONE 3 OF 27

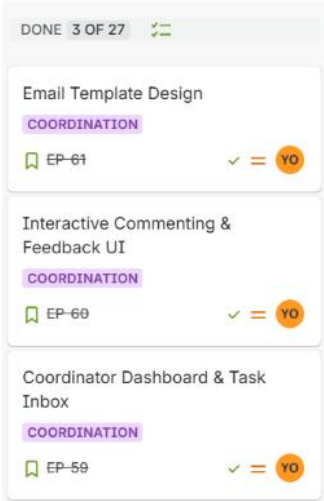
- Automated Email Notification Service (SMTP/SendGrid) — COORDINATION — ~~EP-54~~ — MK
- Faculty-Scoped Article Management Endpoints — COORDINATION — ~~EP-55~~ — MK
- Coordinator Feedback & Commenting API — COORDINATION — ~~EP-53~~ — MK

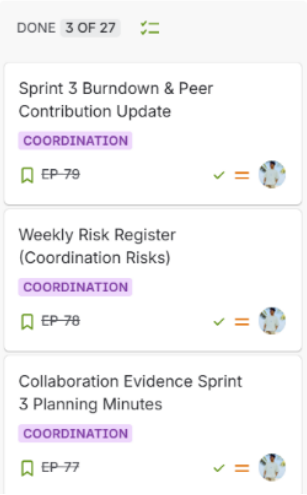
DONE 3 OF 27

- Database Trigger for 14-Day Deadline — COORDINATION — ~~EP-58~~
- Comments & Interaction Schema — COORDINATION — ~~EP-56~~
- Email Log & Notification Audit Trail — COORDINATION — ~~EP-57~~

DONE 3 OF 27

- Overdue Flagging & Priority Indicators — COORDINATION — ~~EP-64~~ — TA
- Multi-Asset Review Panel — COORDINATION — ~~EP-63~~ — TA
- Coordinator Dashboard Layout (Shadcn) — COORDINATION — ~~EP-62~~ — TA


DONE 3 OF 27
Email Template Design
COORDINATION
EP-61
YO
Interactive Commenting & Feedback UI
COORDINATION
EP-60
YO
Coordinator Dashboard & Task Inbox
COORDINATION
EP-59
YO


DONE 3 OF 27
Sprint 3 Burndown & Peer Contribution Update
COORDINATION
EP-79
Weekly Risk Register (Coordination Risks)
COORDINATION
EP-78
Collaboration Evidence Sprint 3 Planning Minutes
COORDINATION
EP-77

In Sprint 4

DONE 3 OF 27

- Evidence of Security Integration (LO2) — ANALYTICS — ~~EP-106~~
- Sprint 4 Velocity Report — ANALYTICS — ~~EP-105~~
- Sprint 4 Planning & Minutes — ANALYTICS — ~~EP-104~~

DONE 3 OF 27

- Marketing Manager Dashboard Wireframes — ANALYTICS — ~~EP-86~~ — YO
- Browser & Usage Analytics Mockups — ANALYTICS — ~~EP-88~~ — YO
- Exception Report UI Design — ANALYTICS — ~~EP-87~~ — YO

DONE 3 OF 27

- Dynamic Exception Report Tables — ANALYTICS — ~~EP-90~~ — TA
- Analytics Dashboard Layout — ANALYTICS — ~~EP-89~~ — TA
- Admin System Settings UI — ANALYTICS — ~~EP-91~~ — TA

DONE 3 OF 27

- Data Sanitization & Backup Script — ANALYTICS — ~~EP-85~~
- Complex Reporting Views — ANALYTICS — ~~EP-83~~
- 14-Day Exception Logic SQL — ANALYTICS — ~~EP-84~~

DONE 3 OF 27

- Exception Reporting Engine — ANALYTICS — ~~EP-81~~ — MK
- V2 Activity & Browser Audit API — ANALYTICS — ~~EP-82~~ — MK
- Analytics Aggregation API — ANALYTICS — ~~EP-80~~ — MK

DONE 6 OF 27

- Final Analytics Dashboard Polish — ANALYTICS — ~~EP-97~~ — HH
- Date Range & Faculty Filters — ANALYTICS — ~~EP-93~~ — HH
- Quality Control & Spec Alignment & Percentage Calculation Logic Verification — ANALYTICS — ~~EP-95~~ — HH

DONE 3 OF 27

Empty State Testing
ANALYTICS
~~EP-100~~

Calculation Accuracy Audit
ANALYTICS
~~EP-98~~

Manager-Role Access Audit
ANALYTICS
~~EP-99~~

DONE 3 OF 27

System Maintenance Guide (Admin)
ANALYTICS
~~EP-103~~

Statistical Analysis Documentation
ANALYTICS
~~EP-101~~

V2 Audit Log Specification
ANALYTICS
~~EP-102~~

In Spring 5

DONE 3 OF 24

- Weighted Scoring Model & Peer Evaluation Setup — SELECTION — ~~EP-128~~
- Burndown Tracking & Velocity Analysis — SELECTION — ~~EP-127~~
- Sprint 5 Ceremony Documentation (Agile Methods) — SELECTION — ~~EP-126~~

DONE 3 OF 24

- Exception Report & Audit Log UI — SELECTION — ~~EP-121~~ — YO
- Responsive Mobile Analytics View — SELECTION — ~~EP-122~~ — YO
- Marketing Manager Dashboard Design (Charts & Stats) — SELECTION — ~~EP-120~~ — YO

DONE 3 OF 24

- Admin V2 Audit Log UI — SELECTION — ~~EP-116~~ — TA
- Dynamic Data Table for Exceptions — SELECTION — ~~EP-115~~ — TA
- Analytics & Reporting Shell — SELECTION — ~~EP-114~~ — TA

DONE 3 OF 24

- Global Statistics Summary Function — SELECTION — ~~EP-131~~
- V2 Requirement: Browser & Device Distribution View — SELECTION — ~~EP-130~~
- Materialized Reporting Views & Performance Indexing — SELECTION — ~~EP-129~~

DONE 3 OF 24

- V2 System Activity & Browser Audit API — SELECTION — ~~EP-110~~ — MK
- Exception Reporting API (Compliance Audit) — SELECTION — ~~EP-109~~ — MK
- Statistical Analysis API (Charts Data) — SELECTION — ~~EP-108~~ — MK

DONE 3 OF 24

- System Usage Visualization (V2 Requirements) — SELECTION — ~~EP-113~~ — HH
- Faculty & Academic Year Filter Logic — SELECTION — ~~EP-112~~ — HH
- Interactive Chart Implementation (Recharts/Chart.js) — SELECTION — ~~EP-111~~ — HH

DONE 3 OF 24

V2 Audit Log & Metadata Injection Test

SELECTION

~~EP-119~~

14-Day Exception & Logic Boundary Test

SELECTION

~~EP-118~~

Analytics Accuracy & Logic Verification

SELECTION

~~EP-117~~

DONE 3 OF 24

UML State Diagram (Contribution Lifecycle)

SELECTION

~~EP-124~~

Statistical Logic & View Documentation

SELECTION

~~EP-125~~

V2 Requirement: System Audit & Tracking Architecture

SELECTION

~~EP-123~~

3.6 Burndown Chart

Burndown charts were drawn in parallel to all sprints to measure the amount of work left in each sprint against the allocated time. By presenting the activities in these graphs, the team was able to see the actual work completed and the remaining tasks. The team used the burndown chart in project implementation for these three purposes.

1. To keep members informed about the daily progress of EchoPress and ensure the sprint is completed on time.

2. To assess whether the product can be implemented by the deadline based on the current development rate.

3. To build the EchoPress platform within the defined scope.

By creating a visual workload chart for each iteration process, the group was able to clearly identify and create a more structured approach to project planning. Burndown charts for activities are shown below.

Sprint 1

Start date: 2 / 23 / 2026.

End date: 3 / 2 / 2026. (Total work days: 5)

Number of works completed: 24

Total work hours completed:

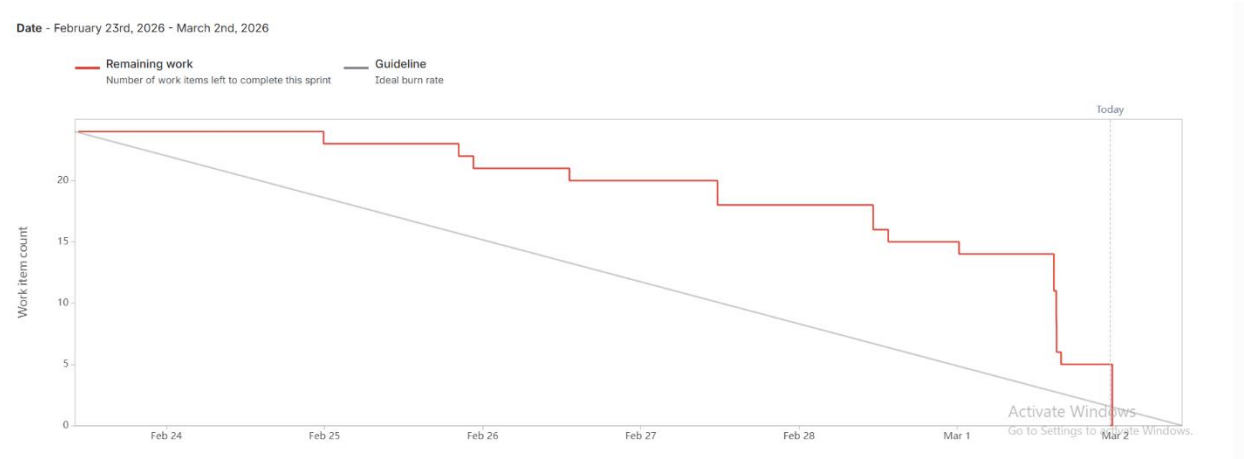


Fig 3.6.1 Burndown chart of Sprint 1

Sprint 2

Start date: 3 / 2 / 2026.

End date: 3 / 9 / 2026. (Total work days: 5)

Number of works completed: 24

Total work hours completed:

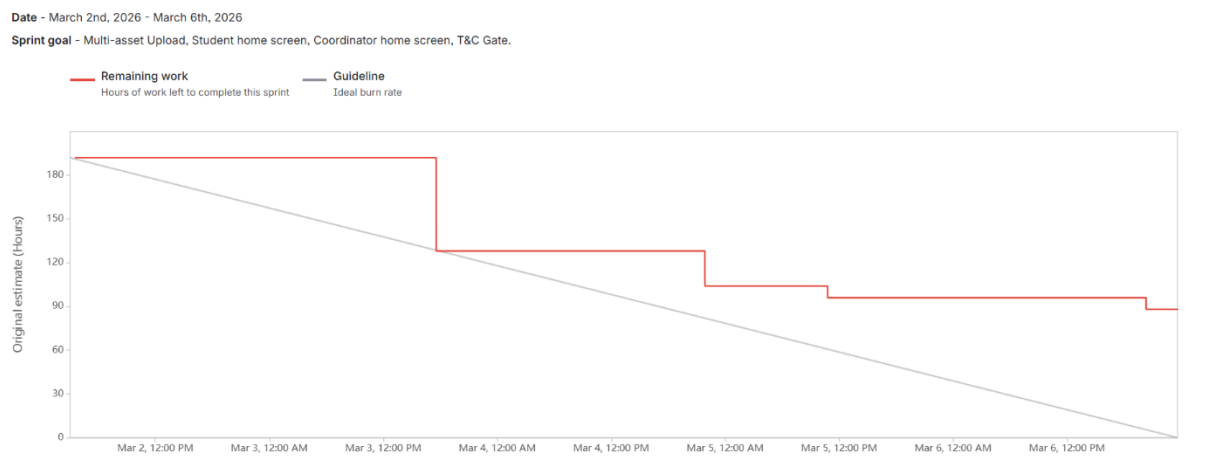


Fig 3.6.2 Burndown chart of Sprint 2.

Sprint 3

Start date: 3 / 9 / 2026.

End date: 3 / 13 / 2026. (Total work days: 5)

Number of works completed: 24

Total work hours completed:

Fig 3.6.3 Burndown chart of Sprint 3.

Sprint 4

Start date: 3 / 16 / 2026.

End date: 3 / 20 / 2026. (Total work days: 5)

Number of works completed: 24

Total work hours completed:

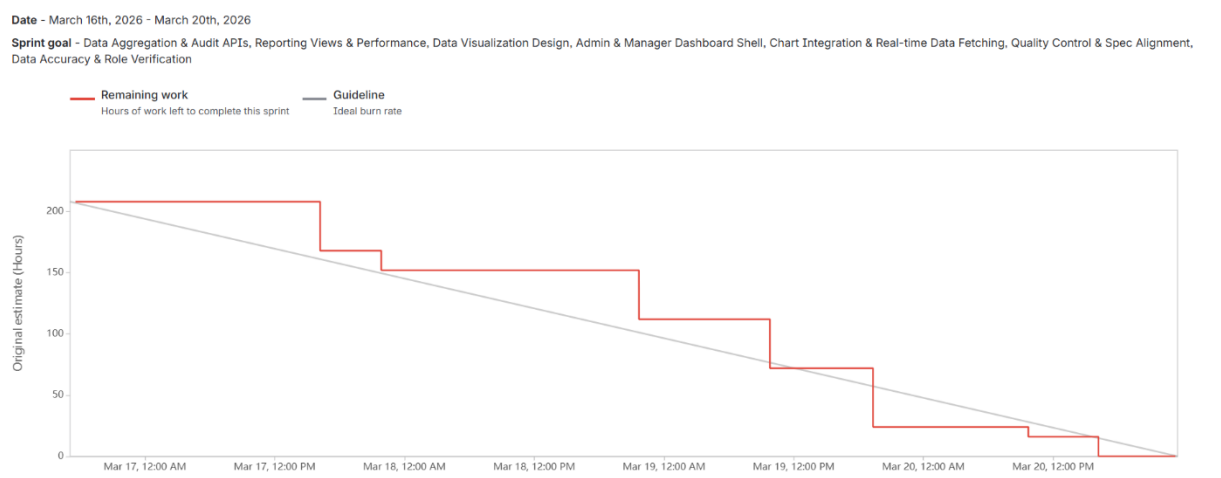


Fig 3.6.4 Burndown chart of Sprint 4.

3.7 Assumption

3.7.1 Permission Management

It is assumed that permissions within the system are managed exclusively by the admin user, ensuring centralized control over access rights. Currently, these access rights are predefined and remain fixed, with no option for modification under the current requirements. However, the system architecture has been deliberately designed with future-proofing in mind, enabling permissions to be adjusted in response to changing requirements or enhancements. (Malsam, 2025)

3.7.2 Roles Management

It is assumed that the system currently defines five distinct roles in accordance with existing requirements, providing structured access and responsibilities. The creation and manipulation of roles are restricted to the admin user, ensuring consistency and centralized governance. Under the current fixed requirements, no new roles will be introduced. Nevertheless, the system has been designed with flexibility, allowing roles to be modified or extended in future iterations to accommodate changing needs. (Malsam, 2025)

3.7.3 Faculties Management

It is assumed that faculty creation and management are restricted to the admin user, ensuring centralized oversight. The system currently defines the following five faculties:

1. Business School.
2. Engineering.
3. Arts & Humanities.
4. Medicine.
5. Science.

Each faculty must have a unique name to prevent duplication and maintain clarity within the system. Those faculties represent the only options available within the current scope of the system, and the total number is limited to five. While new faculties may be added in future iterations to reflect project expansion, the system will not permit the deletion of any faculty that is mapped to existing users, safeguarding data integrity and consistency. (Malsam, 2025)

3.7.4 User management

It is assumed that a single admin user account is predefined and seeded at system initialization, serving as the sole authority for user management. No additional admin accounts can be created, thereby ensuring centralized governance. The admin user is responsible for the creation of all other user accounts, each of which must be assigned a unique login ID and email to maintain system integrity. Users are permitted to change their passwords after the initial setup, providing flexibility in account security. Furthermore, any roles or faculties assigned to users must

already exist within the system, ensuring consistency and alignment with predefined structures. (Malsam, 2025)

User Types

1. **Student:** It is assumed that student accounts are assigned to the student role and are linked to their respective faculty. This ensures that each student's access rights and responsibilities remain consistent with the academic structure of the institution.

2. Coordinator: It is assumed that coordinator accounts are assigned to the coordinator role and are linked to their respective faculty. This structure provides faculty-specific oversight and ensures that coordinators operate within the boundaries of their designated academic unit.

3. Guest: It is assumed that guest accounts are assigned to the guest role and are linked to their respective faculty. This arrangement allows controlled access for external or temporary users while maintaining alignment with faculty-based structures.

4. Manager: It is assumed that manager accounts are assigned to the manager role and may be linked to one or more faculties. Multiple managers may exist to distribute workload across faculties, ensuring balanced oversight. Each manager is responsible for managing the faculties to which they are assigned, maintaining accountability and operational consistency.

5. Admin:

As summary, the assumptions stipulate system operation limits and constraints, as well as guarantee data integrity, security and the management of efficient workflow. The explicitly established roles and tasks facilitate scalability and further development of the system in the future as well, as the system can be adjusted to the changing needs. (Malsam, 2025)

3.8 Database usage and ERD diagram of the system

3.8.1 Justification for Choosing PostgreSQL Database

PostgreSQL was selected as the primary database management system for this project because of its reliability, performance, scalability, and accessibility. The following list highlights the reason behind this choice. (What is PostgreSQL?, n.d.)

1. **Open source and affordable:** It is suitable for student projects and free to use. There are no licensing fees, and the database can be deployed locally or in the cloud with no additional cost.

2. **Free and reliable cloud hosting:** Many platforms such as Neon, Aiven, and Render offer free PostgreSQL cloud hosting (Femi-ige, 2025). The database is currently hosted on the Aiven platform.

3. **Learning and skill Development:** It is widely used in the industry. Working with it helps develop practical skills in SQL, relational modelling and database administration.

4. **High Performance:** Its ability to handle large datasets and complex queries ensures that the system can operate efficiently even under high traffic conditions.

5. **ACID Compliance:** It guarantees reliable transaction processing, maintaining data integrity and consistency.

6. **Scalability**: It can scale efficiently as the number of users increases.

(What is PostgreSQL?, n.d.)

Down below, the ERD diagram of the EchoPress is described.

3.8.2 ERD diagram

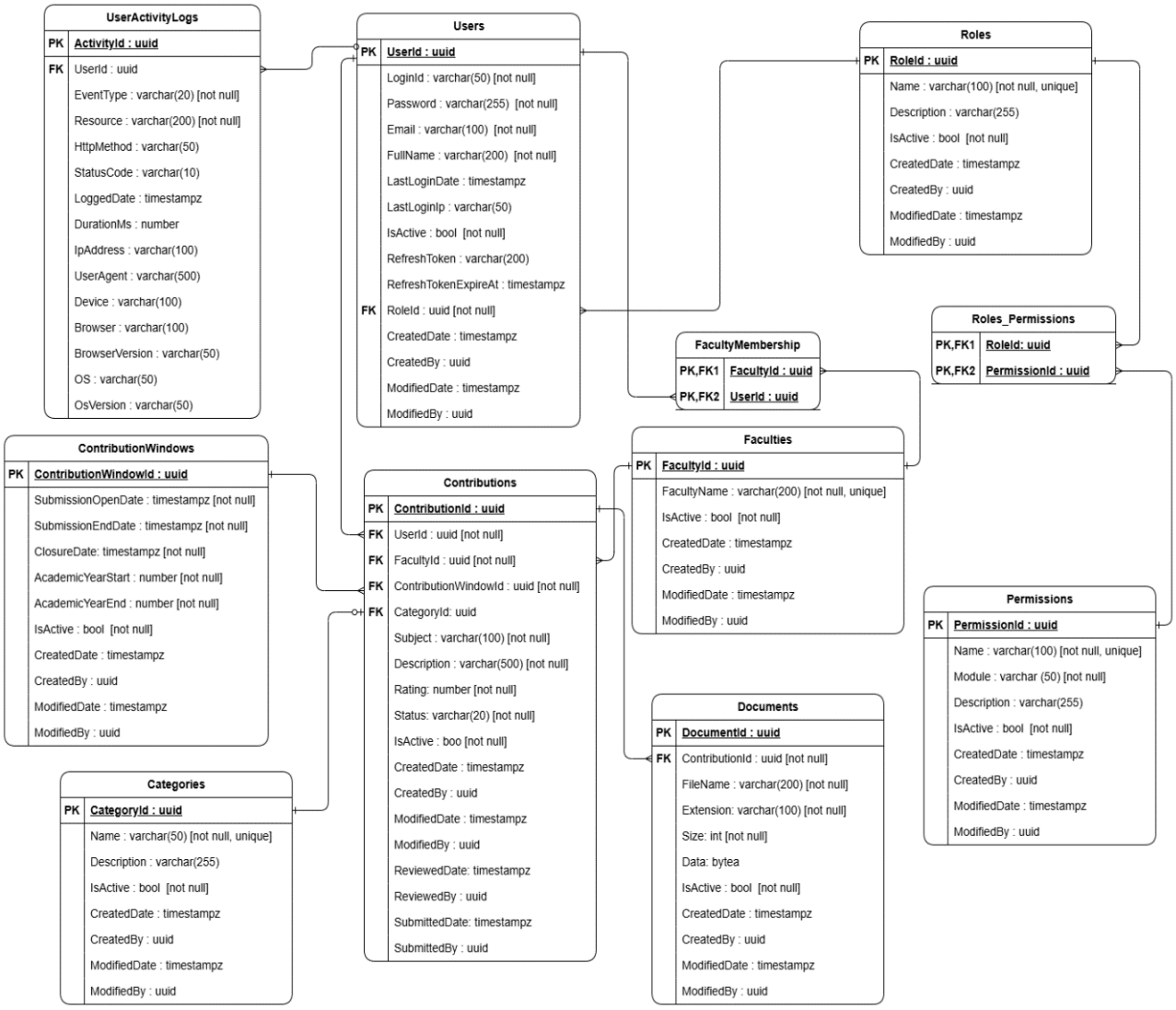


Fig 3.8.1 The ERD diagram of **EchoPress** Platform

3.8.3 Data Dictionaries

Permissions

Column Name	Data Type	Nullable	Default	Description
PermissionId	UUID	No	-	Primary key
Name	VARCHAR (100)	No	-	Unique permission name (e.g., user.create)
Module	VARCHAR (50)	No	-	Module this permission belongs to (e.g., user)
Description	VARCHAR (255)	Yes	-	Description of the permission
IsActive	BOOLEAN	No	TRUE	Indicates if the permission is active
CreatedDate	TIMESTAMPTZ	Yes	-	Record creation timestamp
CreatedBy	UUID	Yes	-	User who created the record
ModifiedDate	TIMESTAMPTZ	Yes	-	Last modification timestamp
ModifiedBy	UUID	Yes	-	User who last modified the record

Constraints: Unique ID (Module, Name).

Roles

Column Name	Data Type	Nullable	Default	Description
RoleId	UUID	No	-	Primary key
Name	VARCHAR (100)	No	-	Unique role name
Description	VARCHAR (255)	Yes	-	Role description
IsActive	BOOLEAN	No	TRUE	Indicates if role is active
CreatedDate	TIMESTAMPTZ	Yes	-	Record creation timestamp
CreatedBy	UUID	Yes	-	Creator
ModifiedDate	TIMESTAMPTZ	Yes	-	Last modification timestamp
ModifiedBy	UUID	Yes	-	Last modifier

Roles Permissions

Column Name	Data Type	Nullable	Default	Description
RoleId	UUID	No	-	Foreign key to Roles
PermissionId	UUID	No	-	Foreign key to Permissions

Constraints: Composite PK (RoleId, PermissionId).

Users

Column Name	Data Type	Nullable	Default	Description
UserId	UUID	No	-	Primary key
LoginId	VARCHAR (50)	No	-	Unique login ID
Password	VARCHAR (255)	No	-	Hashed password
Email	VARCHAR (100)	No	-	Unique email
FullName	VARCHAR (200)	No	-	User full name

LastLoginDate	TIMESTAMPTZ	Yes	-	Last login timestamp (null = first login)
LastLoginIp	VARCHAR (50)	Yes	-	Last login IP
IsActive	BOOLEAN	No	TRUE	User status
RefreshToken	VARCHAR (200)	Yes	-	Refresh token
RefreshTokenExpiresAt	TIMESTAMPTZ	Yes	-	Token expiry
RoleId	UUID	No	-	FK to Roles
CreatedDate	TIMESTAMPTZ	Yes	-	Created timestamp
CreatedBy	UUID	Yes	-	Creator
ModifiedDate	TIMESTAMPTZ	Yes	-	Modified timestamp
ModifiedBy	UUID	Yes	-	Modifier

Faculties

Column Name	Data Type	Nullable	Default	Description
FacultyId	UUID	No	-	Primary key
FacultyName	VARCHAR (200)	No	-	Unique faculty name
IsActive	BOOLEAN	No	TRUE	Active status
CreatedDate	TIMESTAMPTZ	Yes	-	Created timestamp
CreatedBy	UUID	Yes	-	Creator
ModifiedDate	TIMESTAMPTZ	Yes	-	Modified timestamp
ModifiedBy	UUID	Yes	-	Modifier

Faculty Membership

Column Name	Data Type	Nullable	Default	Description
FacultyId	UUID	No	-	FK to Faculties
UserId	UUID	No	-	FK to Users

Constraint: Composite PK (FacultyId, UserId)

Contributions

Column Name	Data Type	Nullable	Default	Description
ContributionId	UUID	No	-	Primary key
UserId	UUID	No	-	FK to Users
FacultyId	UUID	No	-	FK to Faculties
ContributionWindowId	UUID	No	-	FK to ContributionWindows
CategoryId	UUID	Yes	-	FK to Categories
Subject	VARCHAR (100)	No	-	Contribution subject
Description	VARCHAR (500)	No	-	Contribution description
Rating	INT	No	-	Rating (1–10)
Status	VARCHAR (20)	No	-	Contribution status
IsActive	BOOLEAN	No	TRUE	Active status
CreatedDate	TIMESTAMPTZ	Yes	-	Created timestamp
CreatedBy	UUID	Yes	-	Creator
SubmittedDate	TIMESTAMPTZ	Yes	-	Submission date
SubmittedBy	UUID	Yes	-	Submitted by
ReviewedDate	TIMESTAMPTZ	Yes	-	Review date
ReviewedBy	UUID	Yes	-	Reviewer
ModifiedDate	TIMESTAMPTZ	Yes	-	Modified timestamp
ModifiedBy	UUID	Yes	-	Modifier

Categories

Column Name	Data Type	Nullable	Default	Description
CategoryId	UUID	No	-	Primary key
Name	VARCHAR (50)	No	-	Unique category name
Description	VARCHAR (255)	Yes	-	Description
IsActive	BOOLEAN	No	TRUE	Active status
CreatedDate	TIMESTAMPTZ	Yes	-	Created timestamp
CreatedBy	UUID	Yes	-	Creator
ModifiedDate	TIMESTAMPTZ	Yes	-	Modified timestamp
ModifiedBy	UUID	Yes	-	Modifier

Documents

Column Name	Data Type	Nullable	Default	Description
DocumentId	UUID	No	-	Primary key
ContributionId	UUID	No	-	FK to Contributions
FileName	VARCHAR (200)	No	-	Unique file name
Extension	VARCHAR (100)	No	-	File extension
Size	INT	No	-	File size in KB
Data	BYTEA	Yes	-	File binary data
IsActive	BOOLEAN	No	TRUE	Active status
CreatedDate	TIMESTAMPTZ	Yes	-	Created timestamp
CreatedBy	UUID	Yes	-	Creator
ModifiedDate	TIMESTAMPTZ	Yes	-	Modified timestamp
ModifiedBy	UUID	Yes	-	Modifier

Comments

Column Name	Data Type	Nullable	Default	Description
CommentId	UUID	No	-	Primary key
ContributionId	UUID	No	-	FK to Contributions
Message	VARCHAR (500)	No	-	Comment content
IsActive	BOOLEAN	No	TRUE	Active status
CreatedDate	TIMESTAMPTZ	Yes	-	Created timestamp
CreatedBy	UUID	Yes	-	Creator
ModifiedDate	TIMESTAMPTZ	Yes	-	Modified timestamp
ModifiedBy	UUID	Yes	-	Modifier

Contribution Windows

Column Name	Data Type	Nullable	Default	Description
ContributionWindowId	UUID	No	-	Primary key
SubmissionOpenDate	TIMESTAMPTZ	No	-	Start date
SubmissionEndDate	TIMESTAMPTZ	No	-	End date
ClosureDate	TIMESTAMPTZ	No	-	Final closure date
AcademicYearStart	INT	No	-	Academic year start
AcademicYearEnd	INT	No	-	Academic year end
IsActive	BOOLEAN	No	TRUE	Active status
CreatedDate	TIMESTAMPTZ	Yes	-	Created timestamp
CreatedBy	UUID	Yes	-	Creator
ModifiedDate	TIMESTAMPTZ	Yes	-	Modified timestamp
ModifiedBy	UUID	Yes	-	Modifier

UserActivityLogs

Column Name	Data Type	Nullable	Default	Description
ActivityId	UUID	No	-	Primary key
UserId	UUID	Yes	-	FK to Users (null for guest)
EventType	VARCHAR (20)	No	-	Event type (Page_View, API_Call)
Resource	VARCHAR (200)	No	-	Accessed resource
HttpMethod	VARCHAR (50)	Yes	-	HTTP method
StatusCode	VARCHAR (10)	Yes	-	Response status
LoggedDate	TIMESTAMPTZ	Yes	-	Log timestamp
DurationMs	INT	Yes	-	Request duration
IpAddress	VARCHAR (100)	Yes	-	IP address
UserAgent	VARCHAR (500)	Yes	-	User agent
Device	VARCHAR (100)	Yes	-	Device info
Browser	VARCHAR (100)	Yes	-	Browser name
BrowserVersion	VARCHAR (50)	Yes	-	Browser version
OS	VARCHAR (50)	Yes	-	Operating system
OsVersion	VARCHAR (50)	Yes	-	OS version

3.8.4 Data type definitions

This section explains the data types used in the PostgreSQL database schema.

Data Type	Description
UUID	Native PostgreSQL type for storing Universally Unique Identifiers (128-bit). Commonly used as primary keys for distributed systems.
VARCHAR(n)	Variable-length character string with a maximum length of *n*. Enforces length validation at the database level.
BOOLEAN	Stores TRUE, FALSE, or NULL. Used for flags such as IsActive.
INTEGER (INT)	4-byte integer used for whole numbers (e.g., Rating, Year values).
TIMESTAMPTZ	Timestamp with time zone. Stores date and time in UTC internally and converts based on session time zone. Ideal for audit fields.
BYTEA	Binary data type used to store raw byte data such as files or documents.

(Chapter 8. Data Types, n.d.)

3.8.5 Data Normalization

3.9 UseCase diagram

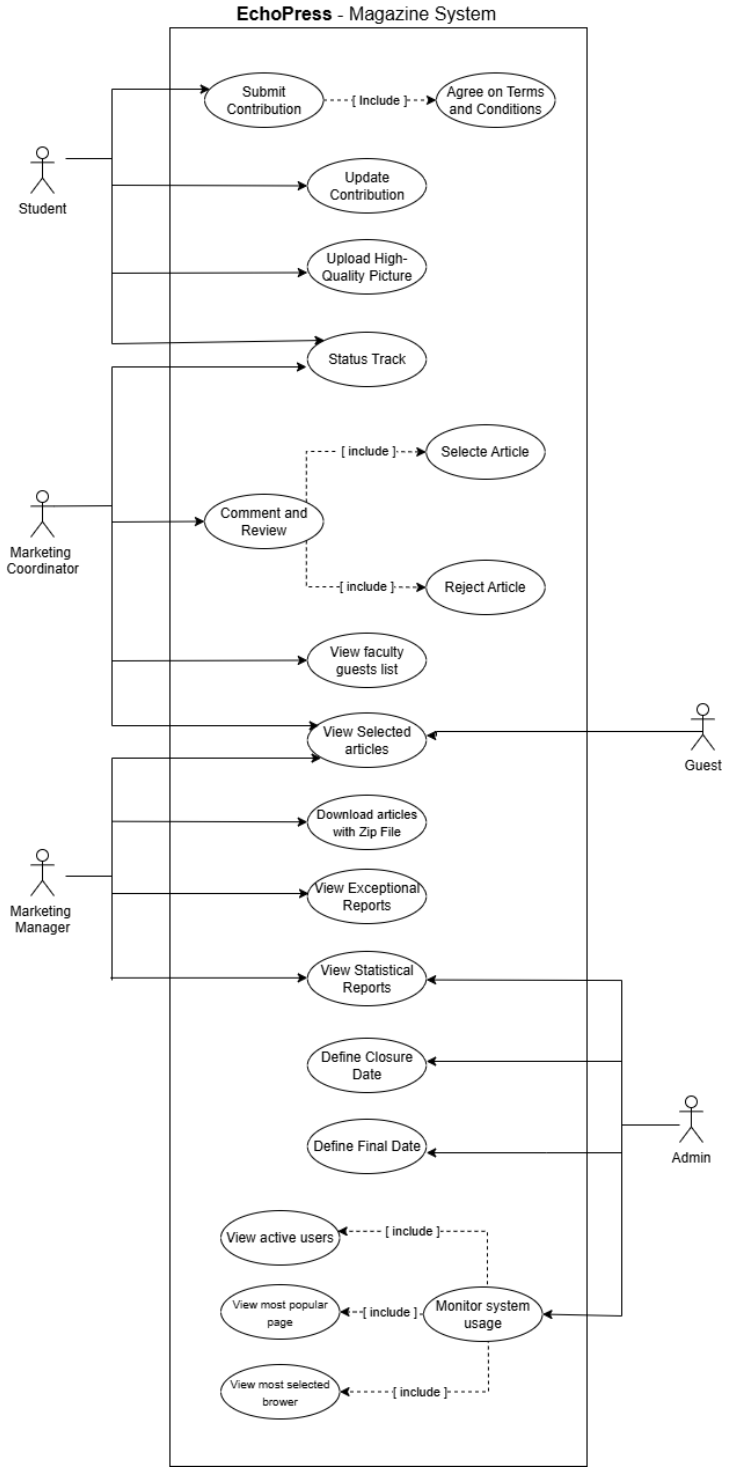


Fig 3.8.2 The UseCase diagram of **EchoPress** Platform

5. Information Architecture

The team created a document structure to make it easier for users to navigate the website and to filter the content within the system. This architecture allowed the group to see how the project's information was organised. Thus, it outlines the organisational structure, design, and sitemap of the system and user responsibilities within system. (Tankala, 2023)

User sorting

The roles of users within the platform are different from one another. Therefore, it is necessary for developers to analyse the content and build it in a user-friendly way. Below are the important points in the content organisation structure of a web system.

1. Users are divided into five distinct groups.

2. Grouping by topic fits the natural mindset of users.

3. Similar grouping across users. (The same pattern even if different, e.g., faculty or browser).

4. A single unified site structure is used to support all the users, eliminating the need to create different sites for each role.

   (Tankala, 2023)

Design Principals

Design principles are established in front-end development to facilitate the user experience within the website. These are:

1. Avoid ambiguous terms and use a formal dictionary.

2. Content is clear and put where people want it.

3. Grouped items are arranged based on the topic.

4. Use consistent elements.

5. Organise elements according to importance.

6. Organise elements according to importance.

Furthermore, the platform is initiated not only to enhance ease of use but also to motivate users to engage with the system. The layout and design structure implemented within the system are in line with modern industry trends, allowing end users to feel comfortable while using the platform and quickly become familiar with its interface. Thus, the theme colour and design structure of the digital portal are briefly outlined below.

(Tankala, 2023)

Theme colour: The colour template of the web platform is based on three colours: red-brown, cream, and white. The team chose these colours because they bring a fresh and modern look to a university website and have a good visual effect on different users. The colour combination is systematically implemented to be suitable for different people. (Tankala, 2023)

Design structure: Hence, the design is a user-centric approach, allowing participants to easily access the platform and carry out relevant processes efficiently. Contributions such as articles and images are displayed in a card-based layout to improve visual attractiveness. Then, user data and system data are presented in tables and charts so that marketing managers, marketing coordinators, and administrators can easily analyse statistical reports. (Tankala, 2023)

Navigation system

A navigation system is included in the web platform that guides participants to the required or desired information without getting lost. It is implemented using API calls. However, since the roles involved vary, it is divided into 2 types: global navigation and role-based navigation.

1. Global navigation: Accessible to any user regardless of role. (System Login).

2. Role-based navigation: Menus will vary depending on the role. This information can be explored further in the sitemap.

   (Tankala, 2023)

Sitemap

A sitemap has been drawn up based on the platform's functionality. A single-site structure was adopted due to the minimal differences between users. Statistical data will be displayed only to key users such as managers, coordinators, and administrators. Student is focused only on article creation and comment communication. Guests are allowed to view only selected articles from the respective faculty. (Tankala, 2023)

While deciding which functions and permissions belong to which user, we found some functionality that was relevant to the system. Top findings are:

1. Students and coordinators are connected through a shared functionality (commenting on articles), which enables communication between them.

2. Login and logout functions are for all the participants.

3. Registered accounts will only be generated by the admin. However, users are able to reset the password of their respective accounts.

   (Tankala, 2023)

Site diagram

Fig 5.1 Site diagram of **EchoPress** platform

The sitemap shown above was used during group project development, allowing Team B1 to better visualise the internal layers of each page in the system and gain a clearer understanding of how pages are interconnected. In addition, this diagram also contributed to effective documentation and system review processes. (Tankala, 2023)

User responsibilities

As mentioned, the roles of users within the platform vary. Therefore, the rights and responsibilities of each role are limited. This restriction is very important for the security and data integrity of the website, as well as for the enhancement of the operation. A brief description of the rights of each role is given below. (Tankala, 2023)

Admin

The Admin has full control over the system and is responsible for:

1. Managing users (**Create, Read, Update, Delete**)
2. Contributing windows **(CRUD operations)** management.
3. Facility management **(CRUD)**
4. Role and permission management to make progressions in the future.
5. Watching analytical reports (that will be determined in subsequent sections)

Coordinator

1. Reviews submitted contributions.
2. Gives feedback on comments.

Have the power to **accept, decline or demand amendments.**

Manager

1. Makes and decides on **approved contributions.**
2. Content management is in charge of higher-level content in faculties.

Student

1. Can create contributions.
2. Can update their own contribution.
3. Can add comments to their own submissions.

Guest

1. Is able to access chosen contributions in his/her assigned faculty.

Can only read and cannot make any changes.

By systematically defining functional duties, it becomes possible to clearly identify who is accountable whenever an error occurs or when a particular process is delayed.

(Tankala, 2023)

In brief, the information architecture document currently describes how structures are organised on the website, how the design is implemented, the blueprint of the entire web platform, and the roles of users within the system. This approach improved development efficiency and team cooperation by reducing time requirements and mitigating the risk of potential logic errors. Furthermore, this architectural document ensures that when new features or content are introduced in the future, the website can be seamlessly integrated with the existing structure.

(Tankala, 2023)

7. Testing of the system

7.1 Introduction

This document outlines the testing methods used to evaluate a website from the user's perspective. The testing focuses only on frontend features and excludes backend or database-level testing. Its purpose is to ensure the website is user-friendly, functional, responsive, and free from visible errors before release.

7.2 Testing Approach

The main method used is Black Box Testing, where the system is tested without knowledge of internal code, focusing only on inputs and outputs.

Additionally, Exploratory Testing is applied to identify unexpected issues through free interaction without predefined test cases.

Testing is conducted manually by simulating real user actions such as clicking, entering data, and navigating pages to ensure correct system behavior.

7.3 Types of testing used for website

7.3.1 Functional testing

Ensures all website features work correctly according to requirements, such as form submissions, button actions, and page navigation.

7.3.2 User Interface (UI) Testing

Checks that all visual elements such as layout, fonts, and buttons are displayed correctly and consistently across pages.

7.3.3 Responsive Testing

Verifies that the website adapts properly to different devices and screen sizes without layout issues.

7.3.4 Cross-Browser Testing

Ensures the website works consistently across browsers like Chrome, Firefox, and Edge.

7.3.5 Validation Testing

Confirms that input fields accept valid data and show appropriate error messages for invalid or empty inputs.

7.3.6 Usability Testing

Evaluates how easy and user-friendly the website is for navigation and task completion.

7.3.7 Regression Testing

Ensures that existing features continue to work correctly after updates or bug fixes.

7.3.8 Exploratory Testing

Allows testers to freely explore the system to discover unexpected bugs through real user behaviour.

7.3.9 Security Testing (Basic)

Checks that unauthorized users cannot access restricted pages and are redirected to the login page.

These testing types are applied through structured test cases in the Test Plan, covering login, CRUD operations, validation, and user management.

7.4 Summary

Frontend testing was conducted using black box and manual testing methods to ensure correct functionality from the user's perspective.

Multiple testing types were applied to evaluate functionality, usability, compatibility, and validation.

Test results show that most cases passed successfully, confirming system stability and usability. One issue (invalid login URL) was found and fixed.

Overall, the website meets requirements and provides a reliable user experience across devices and browsers.

7.5 Test Plan

TC ID	What is being tested	How	Expected results
TC01	Admin login with valid credentials.	Enter valid username and password, click login	System logs in successfully and redirects to dashboard
TC02	Login with empty fields	Click login without entering data.	Validation messages appear
TC03	Login with invalid password.	Enter correct username and wrong password	System shows **Invalid login credentials**
TC04	Login with invalid username.	Enter incorrect username	System shows **Invalid credentials**
TC05	Session persistence.	Login and refresh page	User remains logged in
TC06	Logout.	Click logout button	User is redirected to login page
TC07	View contribution list.	Open contribution page	Contributions display correctly
TC08	Open add contribution form.	Click “Add New”	Form opens successfully
TC9	Add contribution (valid).	Fill all fields and submit	Contribution added successfully
TC10	Add contribution (empty fields).	Submit empty form	Validation messages shown
TC11	Add contribution (long input).	Enter long text	System handles input correctly
TC12	Edit contribution.	Update contribution details	Contribution updated successfully
TC13	Delete contribution.	Click delete button	Contribution removed
TC14	Delete contribution cancel.	Click delete then cancel	Contribution not deleted
TC15	Search contribution (valid).	Enter keyword	Matching results shown
TC16	Search contribution (invalid).	Enter wrong keyword	“No results found” message
TC17	View user list.	Open user management page.	Users displayed correctly

TC18	Open add user form.	Click “Add User”	Form opens
TC19	Add user (valid).	Enter valid details	User created successfully
TC20	Add user (empty fields).	Submit empty form	Validation messages shown
TC21	Add user (invalid email).	Enter incorrect email	Email validation error
TC22	Role selection popup.	Select role in form	Faculty popup appears
TC23	Add duplicate user.	Enter existing username	Duplicate error shown
TC24	Edit user.	Update user details	User updated successfully
TC25	Delete user.	Click delete	User removed
TC26	Delete user cancel.	Cancel delete	User not deleted
TC27	View roles.	Open role management page	Roles displayed
TC28	Open add role form.	Click “Add Role”	Form opens
TC29	Add role (valid).	Enter role name	Role created
TC30	Add role (empty).	Submit empty form	Validation error
TC31	Add duplicate role.	Enter existing role	Duplicate error
TC32	Edit role.	Update role	Role updated
TC33	Delete role.	Click delete	Role removed
TC34	Delete role cancel.	Cancel delete	Role not removed
TC35	View faculty list.	Open faculty page	Faculty displayed
TC36	Open add faculty form.	Click “Add Faculty”	Form opens
TC37	Add faculty (valid).	Enter valid data	Faculty added
TC38	Add faculty (empty).	Submit empty form	Validation error
TC39	Add duplicate faculty.	Enter existing name	Duplicate error
TC40	Edit faculty.	Update details	Faculty updated
TC41	Delete faculty.	Click delete	Faculty removed
TC42	Delete faculty cancel.	Cancel delete	Faculty not removed

TC43	Unauthorized access.	Access without login	Redirect to login
TC44	Invalid login URL.	Enter incorrect URL	System shows 404 Page Not Found

7.6 Test Log

Test	What is being tested	How	Expected Results	Date	Actual results	Action taken
TC01	Admin login with valid credentials	Enter valid username and password, click login	System logs in successfully and redirects to dashboard	23/03/2026	Ok	None
TC02	Login with empty fields	Click login without entering data	Validation messages appear	23/03/2026	Ok	None
TC03	Login with invalid password	Enter correct username and wrong password	System shows Invalid login credentials	23/03/2026	Ok	None
TC04	Login with invalid username	Enter incorrect username	System shows Invalid credentials	23/03/2026	Ok	None
TC05	Session persistence	Login and refresh page	User remains logged in	23/03/2026	Ok	None
TC06	Logout	Click logout button	User is redirected to login page	23/03/2026	Ok	None
TC07	View contribution list	Open contribution page	Contributions display correctly	23/03/2026	Ok	None
TC08	Open add contribution form	Click “Add New”	Form opens successfully	23/03/2026	Ok	None
TC9	Add contribution (valid)	Fill all fields and submit	Contribution added successfully	23/03/2026	Ok	None
TC10	Add contribution (empty fields)	Submit empty form	Validation messages shown	23/03/2026	Ok	None
TC11	Add contribution (long input)	Enter long text	System handles input correctly	23/03/2026	Ok	None
TC12	Edit contribution	Update contribution details	Contribution updated successfully	23/03/2026	Ok	None

TC13	Delete contribution	Click delete button	Contribution removed	23/03/2026	Ok	None
TC14	Delete contribution cancel	Click delete then cancel	Contribution not deleted	23/03/2026	Ok	None
TC15	Search contribution (valid)	Enter keyword	Matching results shown	23/03/2026	Ok	None
TC16	Search contribution (invalid)	Enter wrong keyword	“No results found” message	23/03/2026	Ok	None
TC17	View user list	Open user management page	Users displayed correctly	23/03/2026	Ok	None
TC18	Open add user form	Click “Add User”	Form opens	23/03/2026	Ok	None
TC19	Add user (valid)	Enter valid details	User created successfully	23/03/2026	Ok	None
TC20	Add user (empty fields)	Submit empty form	Validation messages shown	23/03/2026	Ok	None
TC21	Add user (invalid email)	Enter incorrect email	Email validation error	23/03/2026	Ok	None
TC22	Role selection popup	Select role in form	Faculty popup appears	23/03/2026	Ok	None
TC23	Add duplicate user	Enter existing username	Duplicate error shown	23/03/2026	Ok	None
TC24	Edit user	Update user details	User updated successfully	23/03/2026	Ok	None
TC25	Delete user	Click delete	User removed	23/03/2026	Ok	None
TC26	Delete user cancel	Cancel delete	User not deleted	23/03/2026	Ok	None
TC27	View roles	Open role management page	Roles displayed	23/03/2026	Ok	None
TC28	Open add role form	Click “Add Role”	Form opens	23/03/2026	Ok	None
TC29	Add role (valid)	Enter role name	Role created	23/03/2026	Ok	None

TC30	Add role (empty)	Submit empty form	Validation error	23/03/2026	Ok	None
TC31	Add duplicate role	Enter existing role	Duplicate error	23/03/2026	Ok	None
TC32	Edit role	Update role	Role updated	23/03/2026	Ok	None
TC33	Delete role	Click delete	Role removed	23/03/2026	Ok	None
TC34	Delete role cancel	Cancel delete	Role not removed	23/03/2026	Ok	None
TC35	View faculty list	Open faculty page	Faculty displayed	23/03/2026	Ok	None
TC36	Open add faculty form	Click “Add Faculty”	Form opens	23/03/2026	Ok	None
TC37	Add faculty (valid)	Enter valid data	Faculty added	23/03/2026	Ok	None
TC38	Add faculty (empty)	Submit empty form	Validation error	23/03/2026	Ok	None
TC39	Add duplicate faculty	Enter existing name	Duplicate error	23/03/2026	Ok	None
TC40	Edit faculty	Update details	Faculty updated	23/03/2026	Ok	None
TC41	Delete faculty	Click delete	Faculty removed	23/03/2026	Ok	None
TC42	Delete faculty cancel	Cancel delete	Faculty not removed	23/03/2026	Ok	None
TC43	Unauthorized access	Access without login	Redirect to login	23/03/2026	Ok	None
TC44	Invalid login URL	Enter incorrect URL	System shows 404 Page Not Found	23/03/2026	Ok	None